

Muhammad Nadeem

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Professional Summary

AI Research Engineer working on **LLM/VLM and vision models** end to end — agentic RAG, lightweight fine-tuning with Hugging Face PEFT (LoRA/QLoRA), and inference optimization via quantization, ONNX, TensorRT, OpenVINO, and vLLM/Ollama for constrained GPUs and edge devices. M.S. in Computer Science (Sejong University, CVPR Lab); **13 publications** (5 first-author), including WACV 2026 (LLVM-AD workshop) and ACPR 2025; Best Paper Award, ICNGC 2023.

Education

- 2nd Mar-22 **Sejong University**, Seoul, South Korea
18th Feb-24
 - M.S. in Computer Science and Engineering
 - Thesis: FlameNet: Efficient Deep Learning for Fire Detection in Smart Cities
 - GPA: 4.25/4.5, [Presentation Link](#)

1st Feb-14 **The Islamia University of Bahawalpur**, Bahawalpur, Pakistan
23rd Apr-18
 - B.S. in Computer Science
 - GPA: 3.28/4.0

Technical Skills

- LLM / VLM** Qwen2.5-VL, HyperCLOVA X SEED, CLIP dual encoders, VQA, zero-shot cross-modal recognition
- Fine-tuning** Hugging Face Transformers & PEFT, LoRA/QLoRA, instruction tuning, chain-of-thought prompting
- RAG** LangChain, hybrid retrieval (dense + BM25 + knowledge graph), BGE-M3, RRF, cross-encoder reranking
- Inference Opt.** Quantization (INT8/4-bit, PTQ & QAT), ONNX Runtime, TensorRT, OpenVINO, TFLite, distillation
- Serving** vLLM, Ollama, GGUF (llama.cpp), FastAPI, Flask, REST APIs, RTSP ingestion, React dashboards
- Detection** YOLOv8/v12/v26, Faster R-CNN, DETR, SSD, YOLOX, YOLO-NAS, GroundingDINO, MobileNet
- Segmentation** Mask R-CNN instance segmentation, semantic segmentation, OCR pipelines
- Tracking / Pose** Deep SORT, YOLO-pose 17-keypoint skeletons, fall detection, face de-identification
- Generative** FLUX.2 [dev], Stable Diffusion, GANs, diffusers, synthetic data generation
- Training** PyTorch, TensorFlow, OpenCV, multi-GPU (DP, DDP, multi-node), transfer learning
- GPU / Edge HW** NVIDIA GPU servers, Jetson Orin Nano, Raspberry Pi 4/5, Intel N100 edge PC
- Time-Series** N-BEATS, N-HiTS, forecasting and prediction pipelines
- Lang & Tools** Python, C++, Java, C#, Git, JIRA, Agile/Scrum; Ubuntu 24.04, Windows 11, macOS

Selected Projects (Portfolio)

- 2026 **GaN Fab Process-Traceability RAG System** (on-premise LLM + VLM)
Stack: HyperCLOVA X SEED, Qwen2.5-VL 7B, BGE-M3, HF PEFT, Ollama, llama.cpp/GGUF
- Built a fully on-premise Korean RAG assistant over 14,024 chunks from 12,890 equipment records and a 314-page MES manual.
 - Designed hybrid retrieval (BGE-M3 dense search, Korean-bigram BM25, and a PageRank knowledge graph) and fused results with RRF.
 - Added cross-encoder reranking from the top 30 to the top 5 results and returned citations with every answer for auditable use.
 - Used Qwen2.5-VL 7B to caption manual screenshots, turning information embedded in images into searchable text.
 - Domain-adapted the model with LoRA/4-bit QLoRA, merged adapters to GGUF, and served it locally through Ollama.
 - Automated five-step root-cause-analysis and traceability reports; benchmarked candidate LLMs on an in-house RCA evaluation set to select the deployment model.

- 2025 **SpectroFire — Zero-Shot EO/IR Vision-Language Recognition** (ACPR 2025, ICOIN 2026)
- 2026 **Stack:** CLIP-style dual encoder, contrastive learning, EO/IR fusion, PyTorch, Jetson
- Developed a CLIP-style dual encoder that combines electro-optical and thermal-IR imagery for zero-shot fire and smoke recognition.
 - Enabled recognition through cosine similarity between image and text-prompt embeddings, allowing new classes to be added without retraining.
 - Engineered cross-spectral registration, temporal synchronization, and density-based geolocation to align multimodal observations.
 - Prepared the pipeline for Jetson edge inference with automatic CUDA/CPU selection and IoT data networking.
- 2026 **PA-Drive — End-to-End Autonomous Driving Planner** (2026 AD AI Challenge)
- Stack:** Transformer decoder, VAD/VADv2 scene encoder, nuScenes, FP16 autotcast, Docker
- Built a six-camera end-to-end driving planner that converts VAD/VADv2 scene features into K candidate trajectories with a Transformer decoder.
 - Combined a differentiable objective for collision avoidance, lane adherence, and vehicle dynamics with hard-constraint trajectory filtering.
 - Reduced nuScenes planning L2 from **1.902 to 0.246** over four iterations.
 - Corrected temporal data leakage by adopting the official nuScenes splits, improving evaluation validity.
 - Used FP16 autotcast and an explicit FLOPs budget to support on-vehicle latency requirements; related first-author paper accepted at **WACV 2026** (LLVM-AD workshop).
- 2026 **Manhole Worker Safety — Pose-Based Activity Tracking** (ICRA 2027 submission)
- Stack:** YOLOv8/26-pose, OpenCV, RTSP, ONNX Runtime, OpenVINO, FastAPI, Intel N100
- Built an edge safety-monitoring pipeline that tracks 17-keypoint worker poses from live RTSP feeds and detects activities in manholes.
 - Implemented fall detection using torso tilt ($>55^\circ$) or bounding-box aspect ratio (>1.10) sustained for five frames to reduce transient alerts.
 - Designed privacy-by-default de-identification: blur faces in live views, store solid masks, and use shoulder landmarks when face keypoints are unavailable.
 - Logged masking and fall-alert events per frame to provide an auditable record of privacy controls and safety alerts.
 - Fused vision signals with a five-gas sensor array into a four-level edge risk assessment; benchmarked ONNX Runtime and OpenVINO p50/p90/p99 latency on an Intel N100.
- 2026 **Drone Safety-Helmet Compliance Detection with Generative Data Synthesis**
- Stack:** FLUX.2 [dev] via Hugging Face diffusers, YOLO, PyTorch, bfloat16, CPU offload
- Built an aerial safety-helmet compliance detector for construction sites and ski resorts, where labeled drone imagery is limited.
 - Generated image-conditioned synthetic training data with FLUX.2 [dev], using a reference image to retain the intended subject and visual style.
 - Varied backgrounds, weather, and poses in prompts to create 1024×1024 aerial scenes representing ~ 10 m flight altitude.
 - Ran diffusion in bfloat16 with model CPU offload to fit consumer-grade GPUs, then augmented a real YOLO-format helmet dataset with the generated imagery.
- 2024 **Real-Time Detection and Tracking on CCTV and Drone Streams**
- Present **Stack:** YOLOv8/v12/v26, Deep SORT, RTSP/OpenCV, TensorRT, FastAPI, Jetson Orin Nano
- Built reliable real-time inference pipelines for live RTSP feeds from IP/CCTV cameras and drones, including decoding, frame sampling, and automatic reconnection.
 - Delivered detection and multi-object tracking for debris, sea waste, fire/smoke, and people using YOLO and Deep SORT.
 - Improved small-object detection and optimized models with TensorRT for real-time GPU and Jetson edge throughput.
 - Exposed the models as REST APIs and integrated operator-facing visualization dashboards.
- 2022 **FlameNet — Lightweight Fire Detection for Smart Cities** (M.S. Thesis)
- 2024 **Stack:** PyTorch, efficient CNN design, model compression, IoT/edge deployment
- Designed a lightweight CNN for fire detection and classification in IoT environments.
 - Balanced detection accuracy against parameter count and inference cost to make the model suitable for resource-constrained deployment.
 - Published the work in *Smart Cities* (IF 7.0, Q1); related work received the Best Paper Award at ICNGC 2023.

- 2022 **Fake QR Code Recognition with a Lightweight MobileNet Classifier**
Stack: MobileNet, PyTorch, transfer learning, lightweight CNN design
 ○ Developed a classifier that distinguishes authentic from forged QR codes to support counterfeit detection.
 ○ Selected MobileNet's low-parameter architecture to enable phone-based verification instead of server-only inference.
 ○ Applied transfer learning and tuned the accuracy–model-size trade-off for mobile use.
- 2022 **VPBR — Pumpkin Biophysical Property Measurement with Mask R-CNN**
 2023 **Stack:** Mask R-CNN instance segmentation, PyTorch, OpenCV, geometric measurement
 ○ Built a low-cost computer-vision pipeline that segments individual pumpkins with Mask R-CNN and converts masks into length measurements.
 ○ Automated a field-measurement workflow otherwise performed manually, including instance separation under occlusion and varied viewpoints.
 ○ Published as VPBR in *Plants* (IF 4.5, Q1).

Selected Publications

13 publications; 5 as first author. Full list: [Google Scholar](#)

Vision-Language Models, Lightweight Models and Edge Inference (2025–2026)

- WACV 2026 **Lightweight Multi-Scale Fusion for Real-Time Autonomous Driving Segmentation**
Workshop Muhammad Nadeem, Hyeoung Lee
[\[IEEE/CVF WACV 2026 Workshops — 5th Workshop on Large Language and Vision Models for Autonomous Driving \(LLVM-AD\), pp. 1721–1728\]](#)
- ACPR 2025 **Bringing CLIP to the Edge: A Lightweight Fire Detection System for Real-Time Monitoring with UAV**
 Hyeoung Lee, Muhammad Nadeem
[\[Asian Conference on Pattern Recognition \(ACPR\), 2025\]](#)
- ICRCV 2025 **A Label-Free Lightweight Prompt-Driven Cross-Modal Fire Detection on Robotic Edge Platforms**
 Hyeoung Lee, Muhammad Nadeem
[\[The 7th International Conference on Robotics and Computer Vision \(ICRCV\), 2025\]](#)
- ICOIN 2026 **SpectroFire: Zero-Shot Fire and Smoke Detection via Real-Time Edge Inference and IoT Data Networking**
 Hyeoung Lee, Muhammad Nadeem
[\[The 40th International Conference on Information Networking \(ICOIN\), 2026\]](#)
- ICARCA 2025 **EdgeFlame: A Physics-Inspired Framework for Zero-Shot Fire and Smoke Recognition**
 Hyeoung Lee, Muhammad Nadeem
[\[The 4th International Conference on Automation, Robotics and Computer Applications \(ICARCA\), 2025\]](#)

Applied AI, IoT and Signal Processing

- BigComp 2025 **IoT Anomaly Detection with Sound-Based Spectral Analysis and Lightweight Model**
 Hyeoung Lee, Muhammad Nadeem
[\[IEEE International Conference on Big Data and Smart Computing \(BigComp\), 2025\]](#)
- BigComp 2026 **Physics-Consistent Hybrid AI for Quantum Device Manufacturing**
 Hyeoung Lee, Muhammad Nadeem
[\[IEEE International Conference on Big Data and Smart Computing \(BigComp\), 2026\]](#)
- arXiv 2025 **Spectral Feature Extraction for Robust Network Intrusion Detection Using MFCCs**
 Hyeoung Lee, Muhammad Nadeem, Piotr Tsoi
[\[arXiv preprint, 2025\]](#)

Vision and Deep Learning (M.S. Research, 2023)

- Smart Cities **Visual Intelligence in Smart Cities: A Lightweight Deep Learning Model for Fire Detection in an IoT Environment**
IF 7.0, Q1 Muhammad Nadeem, Naqqash Dilshad, Norah Saleh Alghamdi, L Minh Dang, Hyeoung-Kyu Song, Junyoung Nam, Hyeonjoon Moon
[\[Journal, PDF\]](#)

Plants IF 4.5, Q1	VPBR: An Automatic and Low-Cost Vision-Based Biophysical Properties Recognition Pipeline for Pumpkin L. Minh Dang, <u>Muhammad Nadeem</u> , Tan N. Nguyen, Han Yong Park, O New Lee, Hyoung-Kyu Song, Hyeonjoon Moon [Journal, PDF]
ITCCSCC-2023	Machine Learning for Mental Health: A Systematic Study of Seven Approaches for Detecting Mental Disorders <u>Muhammad Nadeem</u> , Junaid Rashid, Hyeonjoon Moon, Arailym Dosset [The 38th International Technical Conference on Circuits/Systems, Computers and Communications, PDF]
ICTC-2023	Wheat Diseases Recognition Using Optimal Features Assisted Modified Soft Attention Network <u>Muhammad Nadeem</u> , Aqib Khan, Ji-Won Kim, L. Minh Dang, Hyeonjoon Moon [The 15th International Conference on ICT Convergence]
ICNGC-2023 Best Paper Award	Brest Cancer Recognition Through Visual Intelligence Assisted Lightweight Convolution Neural Network <u>Muhammad Nadeem</u> , Haseeb Khan, Wisal Khan, L. Minh Dang, Nguyen Le Quan, Hyeonjoon Moon [The 9th International Conference on Next Generation Computing (ICNGC 2023)]

Research and Work Experience

1 st April-2024 Present	AI Research Engineer (LLM/VLM & Computer Vision), Spilab, Seoul, South Korea Tools: Hugging Face, PEFT/LoRA, LangChain, vLLM, ONNX, TensorRT, OpenVINO, YOLO, FastAPI ○ R&D of vision models for detection, segmentation, and tracking on CCTV and drone streams. ○ Built agentic RAG over a 14,000-chunk industrial knowledge base returning cited answers. ○ Hybrid dense + BM25 + knowledge-graph retrieval with cross-encoder reranking. ○ Fine-tuned lightweight LLM/VLMs with Hugging Face PEFT (LoRA/QLoRA) on limited GPU memory. ○ Merged adapters and converted to GGUF for local Ollama serving. ○ Optimized inference via quantization, ONNX, TensorRT, OpenVINO, and vLLM serving. ○ Applied Qwen2.5-VL to document understanding, alongside OCR and VQA pipelines. ○ Delivered debris, sea-waste, fire/smoke, and safety-helmet detection projects. ○ Pose-based worker activity tracking with fall detection and face de-identification. ○ Generated synthetic training data with FLUX.2 [dev] for data-scarce detection tasks. ○ Deployed to Jetson Orin Nano and Raspberry Pi via OpenVINO/TFLite conversion. ○ Built a production multilingual AI chatbot served through FastAPI. ○ Time-series forecasting with N-BEATS and N-HiTS; speech emotion recognition. ○ Published 8 papers (2025–2026), including WACV 2026 workshop and ACPR 2025.
2 nd May-22 31 Jan-24	Graduate Researcher (Computer Vision), CVPR Lab, Sejong University, Seoul, South Korea Tools: PyTorch, Mask R-CNN, YOLO/Faster R-CNN, Stable Diffusion, multi-GPU training, GANs ○ University AI research lab; produced 5 peer-reviewed publications. ○ Designed lightweight CNNs for fire detection, medical imaging, and QR forgery detection. ○ Mask R-CNN pipelines converting masks to physical measurements for plant phenotyping. ○ Scaled training with Data Parallel, DDP, and multi-node GPU clusters. ○ Training, fine-tuning, transfer learning, and evaluation of computer vision models. ○ Built datasets and applied Stable Diffusion and GANs for augmentation. ○ Wrote research papers and technical reports; presented at conferences.
1 st Apr-2018 31 th July-2020	Software Engineer , Systems Junction, Lahore, Pakistan Tools: Java, Android Studio, Retrofit, Room, Glide/Picasso, Firebase, Realm ○ Built and maintained production Android applications in Java. ○ Optimized computer vision algorithms for performance and memory on mobile devices. ○ Developed image processing pipelines and data processing workflows. ○ Worked with MVC and singleton patterns, Firebase, Realm, and push notifications. ○ Mentored junior engineers and maintained technical documentation.

Honors and Achievements

2023	Best Paper Award , ICNGC 2023
2022	Full Tuition Fee Waiver , School of Computer Engineering, Sejong University

2022 **Prof. Stipend**, CVPR Lab, Sejong University
2023 **Conference Travel Grant**, Ministry of Education, South Korea
2014 **Merit Scholarship for Bachelor Degree**, The Islamia University of Bahawalpur, Pakistan

Academic Service

Journal Reviewer: IEEE Trans. on Pattern Analysis and Machine Intelligence (TPAMI), IEEE Trans. on Image Processing (TIP), Fire
Conference Reviewer: 10th Int. Conf. on Next Generation Computing; 40th ITC-CSCC

English Proficiency and Relevant Courseworks

IELTS-AC 6.0 Listening 6.0, Speaking 6.0, Writing 6.0, Reading 5.5

Languages English (4/5), Korean (1/5)

A+ Introduction to Deep Learning, Advanced Deep Learning, Advanced AI

A+ Computer Vision Topics

A0 Advanced HCI

References

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